Five Questions on Fibonacci and Lucas p-Cubes

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Abstract

Wei and Yang introduced Fibonacci and Lucas p-cubes and asked five questions concerning degree distributions, the Wiener index, induced even cycles, farthest vertices, and Hamiltonicity. We give coefficient formulas for the degree distributions of both families, a finite root formula for the Wiener index of Γ_n^p with no summation over the n coordinates, transfer-matrix formulas for induced $2k$-cycles, and dynamic-programming formulas for the number of vertices farthest from a prescribed vertex. For Hamiltonian paths and cycles we record bipartition obstructions, recurrence formulas for these obstructions, small-range consequences, and a necessary-and-sufficient finite integer certificate specialized to Γ_n^p and Λ_n^p. The proofs include the intermediate counting and algebraic steps, so that each formula can be checked directly from the string definitions.

Keywords. Fibonacci cube; Lucas cube; Fibonacci p-cube; Lucas p-cube; degree distribution; Wiener index; induced cycle; eccentricity; Hamiltonian path.

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1 Introduction and notation

Let $[n] = \{1, 2, \ldots, n\}$. For $p \geq 1$, a binary string of length n is a *Fibonacci p-string* if any two of its 1s are separated by at least p consecutive 0s. Equivalently, if the 1s occur in positions $i < j$, then $j - i \geq p + 1$. The subgraph of the hypercube Q_n induced by all Fibonacci p-strings is the *Fibonacci p-cube* Γ_n^p. A *Lucas p-string* is defined by imposing the same separation condition cyclically; the corresponding induced subgraph is the *Lucas p-cube* Λ_n^p. Equivalently, in the Lucas case every cyclic gap between two consecutive selected positions must contain at least p zeros. Thus a one-element support is allowed only when $n > p$, and $\Lambda_n^p \cong K_1$ for $1 \leq n \leq p$.

Classical Fibonacci cubes were introduced by Hsu [6], Lucas cubes by Munarini, Perelli Cippo, and Zagaglia Salvi [14], and their structure is surveyed in [8]. Fibonacci and Lucas p-cubes were introduced and studied in detail by Wei and Yang [15]; their paper contains recursive decompositions, order and size formulas, medianity, distance formulas, the Wiener index formula for Λ_n^p, and cube-polynomial identities. Their concluding section proposed the five questions considered below. For background on partial cubes and median graphs we refer to [5].

Several special or adjacent cases were known before the present work. For $p = 1$, the degree sequences of Fibonacci and Lucas cubes were determined by Klavžar, Mollard, and Petkovšek [12]; the Wiener and Hosoya polynomials were studied by Klavžar and Mollard [10]; short induced cycles in Fibonacci cubes were counted by Eğecioğlu, Saygı, and Saygı [4]; eccentric vertices in Fibonacci and Lucas cubes were studied by Castro and Mollard [2]; and Hamiltonian paths and cycles in ordinary Fibonacci cubes were treated in [3, 16]. After Wei and Yang's paper, Mollard [13] proved their cube-polynomial conjecture for Γ_n^p and determined the distance cube polynomial, but these results do not give the degree distributions, induced-cycle counts, farthest-vertex counts, or Hamiltonian criteria treated here.

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Throughout the paper, $[z^a y^b]P$ denotes the coefficient of the monomial $z^a y^b$ in the polynomial or formal power series P. We put

$$(a)_+ = \max\{a, 0\}.$$

If a binomial coefficient has an invalid lower index or a negative upper index, it is interpreted as zero. The empty string is allowed for Γ_0^p. We write F_t^p for the Fibonacci p-numbers,

$$F_0^p = 0, \qquad F_1^p = \cdots = F_p^p = 1, \qquad F_t^p = F_{t-1}^p + F_{t-p-1}^p \quad (t > p).$$

The vertex sets of Γ_n^p and Λ_n^p are denoted by $\mathcal{F}_n^p$ and $\mathcal{L}_n^p$, respectively. The weight $w(\alpha)$ of a string is the number of 1s in it, and $\mathrm{supp}(\alpha)$ is the set of positions occupied by 1s. Since Γ_n^p and Λ_n^p are partial cubes [15], graph distance is Hamming distance in both families. We use $\mathbf{1}\{\mathcal{E}\}$ for the indicator of an event $\mathcal{E}$.

The following elementary gap notation is used repeatedly. It is included explicitly because most formulas in the paper are obtained by translating a string into its gaps and then recording the length and degree contributions of those gaps.

Lemma 1.1 (Gap encodings)**.** *Let $m \geq 1$.*

1. *A Fibonacci p-string $\alpha \in \mathcal{F}_n^p$ with m ones is uniquely written by the positions*

   $$1 \leq i_1 < i_2 < \cdots < i_m \leq n$$

   and by the gaps

   $$g_0 = i_1 - 1, \qquad g_j = i_{j+1} - i_j - 1 \ (1 \leq j \leq m-1), \qquad g_m = n - i_m.$$

   Here $g_0, g_m \geq 0$, $g_j \geq p$ for $1 \leq j \leq m-1$, and

   $$n = m + \sum_{j=0}^{m} g_j.$$

   Conversely, every such gap vector determines exactly one Fibonacci p-string.

2. *If $n > p$, a nonzero Lucas p-string with m ones and with one of its ones distinguished is encoded by cyclic gaps $h_1, \ldots, h_m$ satisfying*

   $$h_j \geq p \quad (1 \leq j \leq m), \qquad n = m + \sum_{j=1}^{m} h_j.$$

   Each actual Lucas p-string with m ones has exactly m possible distinguished ones.

Proof. For the linear case, g_0 and g_m count the zeros before the first one and after the last one, while g_j counts the zeros between the jth and $(j+1)$st ones. The separation condition says exactly that the internal gaps have length at least p. The total length is the number m of ones plus the total number of zeros, namely $\sum_j g_j$. The reconstruction is obtained by writing

$$0^{g_0} 1 0^{g_1} 1 \cdots 1 0^{g_m}.$$

For the cyclic case, choose one of the m ones as the first one and go around the circle. The cyclic separation condition says that every cyclic gap has length at least p. The same reconstruction works on a circle, and choosing a different one as first gives a different distinguished encoding of the same string. $\square$

2 Degree distribution

This section answers Question 7.1 of Wei and Yang [15]. We begin by recording two gap generating functions. An end gap of length $g \geq 0$ contributes g zeros to the length and $(g-p)_+$ possible insertions of a new 1. Therefore its generating function is

$$\begin{aligned} A_p(z,y) &= \sum_{g\geq 0} z^g y^{(g-p)_+} \\ &= \sum_{0\leq g\leq p} z^g + \sum_{g\geq p+1} z^g y^{g-p} = \frac{1-z^{p+1}}{1-z} + \frac{z^{p+1}y}{1-zy}. \end{aligned}$$

An internal or cyclic gap has length $g \geq p$ and contributes $(g-2p)_+$ insertions; hence

$$\begin{aligned} B_p(z,y) &= \sum_{g\geq p} z^g y^{(g-2p)_+} \\ &= \sum_{p\leq g\leq 2p} z^g + \sum_{g\geq 2p+1} z^g y^{g-2p} = \frac{z^p - z^{2p+1}}{1-z} + \frac{z^{2p+1}y}{1-zy}. \end{aligned}$$

Here z records the number of zeros in the gap, while y records the number of possible insertions into that gap.

Theorem 2.1 (Degree distribution of Γ_n^p)**.** *Let $N_\Gamma^p(n,k)$ be the number of vertices of degree k in Γ_n^p. Then*

$$N_\Gamma^p(n,k) = \mathbf{1}\{k=n\} + \sum_{m\geq 1} [z^{n-m}y^{k-m}] A_p(z,y)^2 B_p(z,y)^{m-1}.$$

Equivalently,

$$\sum_{n\geq 0}\sum_{k\geq 0} N_\Gamma^p(n,k) z^n y^k = \frac{1}{1-zy} + \frac{zyA_p(z,y)^2}{1-zyB_p(z,y)}.$$

Proof. We separate the all-zero vertex from the nonzero vertices. In Γ_n^p, the all-zero vertex is adjacent to all n strings with exactly one 1, so it has degree n. Its contribution to the bivariate generating function is therefore

$$\sum_{n\geq 0} z^n y^n = \frac{1}{1-zy}.$$

Now let α be a nonzero vertex with $m \geq 1$ ones, and use the gaps from theorem 1.1. There are two types of neighbours of α.

First, every existing 1 can be changed to 0. This never creates two ones that are too close, so it always gives a vertex of Γ_n^p. Hence deletions contribute exactly m to the degree. In a generating function this accounts for a factor $z^m y^m$: the factor z^m records the m positions occupied by ones, and the factor y^m records the m deletion neighbours.

Second, a zero can be changed to 1 only if the new 1 is at distance at least $p+1$ from every old 1. In the left end gap of length g_0, the allowed zeros are those at least p zeros away from the first existing 1, so there are $(g_0-p)_+$ choices. The same gives $(g_m-p)_+$ choices in the right end gap. In an internal gap of length g_j between two old ones, the new 1 must leave at least p zeros on both sides; hence the number of choices is $(g_j-2p)_+$. Thus

$$\deg_{\Gamma_n^p}(\alpha) = m + (g_0-p)_+ + (g_m-p)_+ + \sum_{j=1}^{m-1}(g_j-2p)_+.$$

For fixed m, the two end gaps are recorded by $A_p(z,y)^2$, and the $m-1$ internal gaps are recorded by $B_p(z,y)^{m-1}$. Since the m ones already contribute length m and degree m, the coefficient of

$z^{n-m}y^{k-m}$ in $A_p(z,y)^2B_p(z,y)^{m-1}$ counts exactly the vertices of length n and degree k with m ones. Summing over $m \geq 1$ proves the coefficient formula.

For the generating-function form, sum the fixed-m contribution:

$$\sum_{m\geq 1} z^m y^m A_p(z,y)^2 B_p(z,y)^{m-1} = zyA_p(z,y)^2 \sum_{m\geq 1} (zyB_p(z,y))^{m-1} = \frac{zyA_p(z,y)^2}{1 - zyB_p(z,y)}.$$

Adding the all-zero contribution gives the stated identity. □

Theorem 2.2 (Degree distribution of Λ_n^p)**.** *Let $N_\Lambda^p(n,k)$ be the number of vertices of degree k in Λ_n^p. If $1 \leq n \leq p$, then*

$$N_\Lambda^p(n,k) = \mathbf{1}\{k = 0\}.$$

If $n > p$, then

$$N_\Lambda^p(n,k) = \mathbf{1}\{k = n\} + \sum_{m\geq 1} \frac{n}{m}[z^{n-m}y^{k-m}]B_p(z,y)^m.$$

Proof. For $n \leq p$, $\Lambda_n^p \cong K_1$ by [15, Proposition 2.5], so the unique vertex has degree 0.

Assume $n > p$. The all-zero vertex has degree n, since changing any one coordinate to 1 produces a Lucas p-string with a single 1. Now consider a nonzero Lucas p-string with m ones. Choose one of the m ones as distinguished and let $h_1, \ldots, h_m$ be the cyclic gaps after successive ones. By theorem 1.1, each $h_j \geq p$ and $n = m + \sum_j h_j$.

As in the linear case, deleting any of the m existing ones gives a neighbour. Insertions are now possible only inside cyclic gaps. In a cyclic gap of length h_j, a new 1 must leave at least p zeros from the preceding old 1 and at least p zeros from the following old 1, so it contributes $(h_j - 2p)_+$ insertion neighbours. Hence a distinguished cyclic gap vector contributes

$$z^m y^m B_p(z,y)^m.$$

The coefficient $[z^{n-m}y^{k-m}]B_p(z,y)^m$ therefore counts distinguished gap vectors of length n and degree k after the m ones and m deletion neighbours have been accounted for.

There are n possible positions for the distinguished first 1. However, each actual string with m ones is obtained m times, once for each possible distinguished one. Thus we multiply by n/m. This proves the formula, together with the separate all-zero term. □

Example 2.3. For $p = 2$ and $n = 4$, the vertices of Γ_4^p are

$$0000,\ 1000,\ 0100,\ 0010,\ 0001,\ 1001.$$

Their degrees are respectively

$$4,\ 2,\ 1,\ 1,\ 2,\ 2.$$

Thus $N_\Gamma^2(4,1) = 2$, $N_\Gamma^2(4,2) = 3$, and $N_\Gamma^2(4,4) = 1$. This small example illustrates the two operations used in the proof: deleting an existing 1, and inserting a new 1 only when at least two zeros remain between any two ones.

3 A root formula for the Wiener index of Γ_n^p

This section answers Question 7.2 in the sense of a finite formula depending only on the roots of the characteristic polynomial. We first spell out the semicube calculation from which the root formula starts. Recall that, for a partial cube isometrically embedded into a hypercube, the Wiener index is the sum over coordinates of the product of the two semicube sizes [9].

Lemma 3.1 (Semicube counts for Γ_n^p)**.** *For $1 \le i \le n$, let*

$$S_i = \{\alpha \in \mathcal{F}_n^p : \alpha_i = 1\}.$$

Then

$$|S_i| = F_i^p F_{n-i+1}^p.$$

Consequently,

$$W(\Gamma_n^p) = F_{n+p+1}^p \sum_{i=1}^{n} F_i^p F_{n-i+1}^p - \sum_{i=1}^{n} (F_i^p)^2 (F_{n-i+1}^p)^2.$$

Proof. Fix a coordinate i and require $\alpha_i = 1$. Then the p positions immediately to the left of i, if they exist, and the p positions immediately to the right of i, if they exist, must be zero. The coordinates to the left of this forced block form an arbitrary Fibonacci p-string, and the coordinates to the right form another arbitrary Fibonacci p-string. More explicitly, the free left part has length $(i-p-1)_+$. Its number of choices is F_i^p: if $i \le p+1$ this number is $1 = F_i^p$, while if $i \ge p+2$ it is

$$|\mathcal{F}_{i-p-1}^p| = F_{(i-p-1)+p+1}^p = F_i^p$$

by the order formula of Wei and Yang [15, Theorem 4.1]. The same argument on the right gives F_{n-i+1}^p choices. Hence $|S_i| = F_i^p F_{n-i+1}^p$.

Since Γ_n^p is a partial cube, graph distance equals Hamming distance and the semicube formula gives

$$W(\Gamma_n^p) = \sum_{i=1}^{n} |S_i| \big(|\mathcal{F}_n^p| - |S_i| \big).$$

Using $|\mathcal{F}_n^p| = F_{n+p+1}^p$ and substituting the value of $|S_i|$ gives the displayed formula. □

Now let

$$q_p(x) = x^{p+1} - x^p - 1,$$

and let $\rho_1, \dots, \rho_{p+1}$ be its roots. These roots are simple: if a root were also a root of

$$q_p'(x) = (p+1)x^p - p x^{p-1} = x^{p-1}((p+1)x - p),$$

then it would have to be 0 or $p/(p+1)$, but neither value satisfies $q_p(x) = 0$. Put

$$c_j = \frac{1}{(p+1)\rho_j - p} \qquad (1 \le j \le p+1).$$

Lemma 3.2 (Binet expansion)**.** *For every $t \ge 0$,*

$$F_t^p = \sum_{j=1}^{p+1} c_j \rho_j^t.$$

Proof. The generating function of the sequence F_t^p is

$$\sum_{t \ge 0} F_t^p x^t = \frac{x}{1 - x - x^{p+1}}.$$

Since $q_p(1/x) = x^{-(p+1)}(1 - x - x^{p+1})$, we have

$$1 - x - x^{p+1} = \prod_{j=1}^{p+1} (1 - \rho_j x).$$

Therefore

$$\frac{x}{1-x-x^{p+1}} = \sum_{j=1}^{p+1} \frac{A_j}{1-\rho_j x}$$

for suitable constants A_j. Multiplying by $1-\rho_j x$ and setting $x = 1/\rho_j$ gives

$$A_j = \frac{1/\rho_j}{\prod_{r\neq j}(1-\rho_r/\rho_j)}.$$

Because

$$q_p'(\rho_j) = \prod_{r\neq j}(\rho_j - \rho_r) = \rho_j^p \prod_{r\neq j}(1-\rho_r/\rho_j),$$

we get

$$A_j = \frac{\rho_j^{p-1}}{q_p'(\rho_j)} = \frac{\rho_j^{p-1}}{\rho_j^{p-1}((p+1)\rho_j - p)} = \frac{1}{(p+1)\rho_j - p} = c_j.$$

Finally, $1/(1-\rho_j x) = \sum_{t\geq 0} \rho_j^t x^t$, and comparing coefficients gives the formula. □

For complex numbers a, b, define

$$G_n(a,b) = \begin{cases} \dfrac{ab(a^n - b^n)}{a-b}, & a \neq b, \\ na^{n+1}, & a = b. \end{cases}$$

This notation records the finite geometric identity

$$G_n(a,b) = \sum_{i=1}^{n} a^i b^{n-i+1}.$$

Indeed, if $a \neq b$, then

$$\sum_{i=1}^{n} a^i b^{n-i+1} = ab \sum_{j=0}^{n-1} a^j b^{n-1-j} = ab\frac{a^n - b^n}{a-b},$$

and if $a = b$, all n terms are equal to a^{n+1}.

Theorem 3.3 (Root formula for $W(\Gamma_n^p)$)**.** *For $p \geq 1$ and $n \geq 1$,*

$$W(\Gamma_n^p) = \left(\sum_{\ell=1}^{p+1} c_\ell \rho_\ell^{n+p+1}\right)\left(\sum_{r,s=1}^{p+1} c_r c_s G_n(\rho_r, \rho_s)\right) - \sum_{r,s,u,v=1}^{p+1} c_r c_s c_u c_v G_n(\rho_r\rho_s, \rho_u\rho_v).$$

Proof. Start from theorem 3.1. The first factor in the first term is

$$F_{n+p+1}^p = \sum_{\ell=1}^{p+1} c_\ell \rho_\ell^{n+p+1}$$

by theorem 3.2. For the inner sum we substitute the same expansion twice:

$$\sum_{i=1}^{n} F_i^p F_{n-i+1}^p = \sum_{i=1}^{n}\left(\sum_{r=1}^{p+1} c_r \rho_r^i\right)\left(\sum_{s=1}^{p+1} c_s \rho_s^{n-i+1}\right)$$
$$= \sum_{r,s=1}^{p+1} c_r c_s \sum_{i=1}^{n} \rho_r^i \rho_s^{n-i+1} = \sum_{r,s=1}^{p+1} c_r c_s G_n(\rho_r, \rho_s).$$

For the second term in theorem 3.1, expand four copies of the Binet formula:

$$\begin{aligned}\sum_{i=1}^{n}(F_i^p)^2(F_{n-i+1}^p)^2 &= \sum_{r,s,u,v=1}^{p+1} c_r c_s c_u c_v \sum_{i=1}^{n}(\rho_r\rho_s)^i(\rho_u\rho_v)^{n-i+1} \\ &= \sum_{r,s,u,v=1}^{p+1} c_r c_s c_u c_v G_n(\rho_r\rho_s,\rho_u\rho_v).\end{aligned}$$

Substitution gives the stated formula. Although complex roots occur, the final value is real because it is the Wiener index of a finite graph. □

4 Induced even cycles

This section gives an exact transfer-matrix answer to Question 7.3. Related enumerations of short induced cycles in ordinary Fibonacci cubes appear in [7, 4].

Fix $q = 2k \geq 4$. We want to count ordered q-tuples of vertices that form an induced cycle in the cyclic order

$$v^0, v^1, \dots, v^{q-1}, v^0.$$

The transfer matrix below constructs the q strings coordinate by coordinate. Let $\mathcal{S}_{p,q}$ be the set of all $q \times p$ binary matrices in which every row contains at most one 1. Thus a state stores the last p columns already constructed. If $\sigma = (s_1, \dots, s_p) \in \mathcal{S}_{p,q}$, where each $s_i \in \{0,1\}^q$ is a column, and if $b = (b_0, \dots, b_{q-1}) \in \{0,1\}^q$, the transition obtained by appending b is allowed when every row in which b has a 1 is zero throughout σ. This is exactly the condition that a new 1 is not placed within the previous p positions of an old 1 in the same row. The new state is

$$\tau = (s_2, \dots, s_p, b).$$

Introduce variables x_{rs} for $0 \leq r < s < q$ and assign the transition weight

$$\omega(b) = \prod_{0 \leq r < s < q} x_{rs}^{|b_r - b_s|}.$$

At one coordinate, the factor x_{rs} appears exactly when the rth and sth strings differ in that coordinate. After multiplying all transition weights, the exponent of x_{rs} is therefore the Hamming distance between the two completed strings.

Let $M_{p,q}(x)$ be the $\mathcal{S}_{p,q} \times \mathcal{S}_{p,q}$ matrix whose (σ,τ) entry is $\omega(b)$ if the transition $\sigma \to \tau$ is obtained by appending b, and is 0 otherwise. Let σ_0 be the all-zero state, let e_0 be its coordinate vector, and let $\mathbf{1}$ be the all-one column vector. Define

$$P_\Gamma(n,p,q;x) = e_0^\top M_{p,q}(x)^n \mathbf{1}.$$

The initial zero state means that the first p nonexistent positions before coordinate 1 are treated as zeros, and the final multiplication by $\mathbf{1}$ means that no extra condition is imposed after coordinate n.

For the cyclic Lucas condition, when $n > p$ define

$$P_\Lambda(n,p,q;x) = \operatorname{tr}\big(M_{p,q}(x)^n\big).$$

Taking the trace forces the final state to equal the initial state, which imposes the wrap-around separation conditions. For $n \leq p$, $\Lambda_n^p \cong K_1$, so it contains no cycles.

Let

$$E_q = \{\{r, r+1\} : 0 \leq r < q\},$$

where indices are taken modulo q, and let N_q be the set of all unordered nonconsecutive pairs in $\{0, \ldots, q-1\}$. For a polynomial P in the variables x_{rs}, write

$$[C_q]P = \sum_{\substack{e_{rs} \geq 0:\ e_{rs}=1 \text{ for } \{r,s\} \in E_q, \\ e_{rs} \geq 2 \text{ for } \{r,s\} \in N_q}} \left[\prod_{0 \leq r < s < q} x_{rs}^{e_{rs}} \right] P.$$

Thus $[C_q]P$ retains exactly those ordered q-tuples in which consecutive vertices have Hamming distance 1 and nonconsecutive vertices have Hamming distance at least 2.

Theorem 4.1 (Induced $2k$-cycles)**.** *For $k \geq 2$,*

$$C_{2k}(\Gamma_n^p) = \frac{1}{4k}[C_{2k}]P_\Gamma(n, p, 2k; x).$$

For $n > p$,

$$C_{2k}(\Lambda_n^p) = \frac{1}{4k}[C_{2k}]P_\Lambda(n, p, 2k; x),$$

and $C_{2k}(\Lambda_n^p) = 0$ for $n \leq p$.

Proof. The preceding construction produces ordered q-tuples $(v^0, \ldots, v^{q-1})$ of Fibonacci p-strings in the linear case and Lucas p-strings in the cyclic case. For each pair $r < s$, the exponent of x_{rs} in the product of transition weights is

$$\sum_{j=1}^{n} |v_j^r - v_j^s| = H(v^r, v^s).$$

Since Γ_n^p and Λ_n^p are induced subgraphs of the hypercube Q_n, two vertices are adjacent exactly when their Hamming distance is 1.

Therefore an ordered tuple is an induced q-cycle in the displayed cyclic order exactly when the q consecutive pairs have Hamming distance 1 and every nonconsecutive pair has Hamming distance at least 2. If a nonconsecutive pair had distance 1, it would be a chord; if it had distance 0, two vertices of the tuple would coincide. The operator $[C_q]$ extracts exactly these tuples.

An undirected induced q-cycle has q choices of starting vertex and 2 choices of orientation. Hence it is counted $2q = 4k$ times among the ordered tuples. Dividing by $4k$ gives the result. ☐

5 Farthest vertices from a prescribed vertex

This section answers Question 7.4. For the case $p = 1$, eccentric vertices in Fibonacci and Lucas cubes were studied by Castro and Mollard [2]. Let μ be a vertex of Γ_n^p or Λ_n^p, and put

$$Z(\mu) = \{i : \mu_i = 0\}.$$

A set $S \subseteq [n]$ is *linearly p-separated* if $|i - j| \geq p + 1$ for all distinct $i, j \in S$. It is *cyclically p-separated* if the corresponding cyclic gaps between consecutive selected positions each contain at least p zeros; the empty set is admissible, and a singleton is cyclically admissible precisely when $n > p$.

Lemma 5.1. *Let X be either Γ_n^p or Λ_n^p. If ν is farthest from μ in X, then*

$$\mathrm{supp}(\nu) \cap \mathrm{supp}(\mu) = \varnothing.$$

Consequently, farthest vertices from μ are exactly the vertices whose supports are maximum-cardinality admissible subsets of $Z(\mu)$, with the linear separation rule for Γ_n^p and the cyclic separation rule for Λ_n^p.

Proof. Suppose, to the contrary, that there is a coordinate i with $\mu_i = \nu_i = 1$. Change the ith bit of ν from 1 to 0 and call the new string ν'. Deleting a 1 cannot create a forbidden pair of ones, so ν' is still a vertex of X. At coordinate i, the contribution to the Hamming distance from μ changes from 0 to 1, while all other coordinates are unchanged. Hence

$$d(\mu, \nu') = d(\mu, \nu) + 1,$$

contradicting the assumption that ν is farthest from μ.

Thus a farthest ν has support contained in $Z(\mu)$. For every such ν,

$$d(\mu, \nu) = w(\mu) + w(\nu),$$

because every 1 of μ contributes one differing coordinate and every selected position of ν contributes one more. The fixed term $w(\mu)$ does not depend on ν, so maximizing the distance is the same as maximizing the size of an admissible support contained in $Z(\mu)$. □

Theorem 5.2 (Linear dynamic program for Γ_n^p)**.** *Let $a_i = 1 - \mu_i$. Thus $a_i = 1$ exactly when position i is available for the support of a farthest vertex. Define M_i, C_i for $i \le 0$ by $M_i = 0$ and $C_i = 1$. For $1 \le i \le n$, if $a_i = 0$, put*

$$M_i = M_{i-1}, \qquad C_i = C_{i-1}.$$

If $a_i = 1$, put

$$u_i = M_{i-1}, \qquad v_i = 1 + M_{i-p-1}, \qquad M_i = \max\{u_i, v_i\},$$

and update the number of optimal choices by

$$C_i = \mathbf{1}\{u_i = M_i\}C_{i-1} + \mathbf{1}\{v_i = M_i\}C_{i-p-1}.$$

Then

$$e_{\Gamma_n^p}(\mu) = w(\mu) + M_n,$$

and the number of vertices ν satisfying $d(\mu, \nu) = e(\mu)$ is C_n.

Proof. For $i \ge 0$, let M_i be the maximum size of a linearly p-separated subset of $Z(\mu) \cap \{1, \dots, i\}$, and let C_i be the number of subsets attaining this maximum. For $i \le 0$, there is only the empty subset, so the initial values $M_i = 0$ and $C_i = 1$ are correct.

Now consider position i. If $a_i = 0$, then $i \notin Z(\mu)$ and it cannot be selected; every optimal subset of $\{1, \dots, i\}$ is exactly an optimal subset of $\{1, \dots, i-1\}$. Hence $M_i = M_{i-1}$ and $C_i = C_{i-1}$.

Assume $a_i = 1$. There are two disjoint classes of admissible subsets.

- If i is not selected, the best possible size is $u_i = M_{i-1}$, with C_{i-1} choices.
- If i is selected, then no selected position may lie in $\{i-p, \dots, i-1\}$. The remaining selected positions must lie in $\{1, \dots, i-p-1\}$, so the best possible size is $v_i = 1 + M_{i-p-1}$, with C_{i-p-1} choices.

Therefore $M_i = \max\{u_i, v_i\}$. If only one of u_i, v_i equals this maximum, we keep only the corresponding choices; if they are equal, both disjoint classes are optimal and their counts add. This is exactly the displayed formula for C_i.

By theorem 5.1, farthest vertices from μ are obtained by maximum linearly p-separated subsets of $Z(\mu)$. Thus the maximum extra contribution is M_n, the number of farthest vertices is C_n, and the eccentricity is $w(\mu) + M_n$. □

Remark 5.3. The symbols u_i and v_i in theorem 5.2 have a simple meaning: u_i is the best value when the last position is skipped, and v_i is the best value when the last position is used. This is the same "skip or take" dynamic-programming principle used for independent sets in paths, except that the forbidden window has length p rather than length 1.

Theorem 5.4 (Cyclic dynamic program for Λ_n^p)**.** *If $n \le p$, then $\Lambda_n^p \cong K_1$, $e_{\Lambda_n^p}(\mu) = 0$, and the number of farthest vertices is 1. Assume $n > p$. Let $\mathcal{S}_{p,1}$ be the one-row state set used in Section 4. For each coordinate i, define a transfer matrix $T_i(t)$ on $\mathcal{S}_{p,1}$ as follows: the transition appending $b \in \{0,1\}$ is allowed if $b \le 1 - \mu_i$ and, when $b = 1$, the current state is the zero state; its weight is t^b. Put*

$$P_\mu^\circ(t) = \operatorname{tr}(T_1(t)T_2(t)\cdots T_n(t)).$$

If

$$M^\circ = \deg P_\mu^\circ(t), \qquad C^\circ = [t^{M^\circ}]P_\mu^\circ(t),$$

then

$$e_{\Lambda_n^p}(\mu) = w(\mu) + M^\circ,$$

and the number of farthest vertices from μ is C°.

Proof. A state records the last p decisions in the cyclic construction of a subset of $Z(\mu)$. The condition $b \le 1 - \mu_i$ says that we may select position i only if $\mu_i = 0$. If $b = 1$, the current state must be all zero, which says that no selected position occurs among the previous p positions. Thus every product of transitions records a p-separated choice of positions.

The trace imposes the equality of the final and initial states. This is precisely what enforces the separation condition across the boundary between coordinates n and 1. Therefore $P_\mu^\circ(t)$ is the independence polynomial of cyclically p-separated subsets of $Z(\mu)$, where the exponent of t is the subset size. Its degree M° is the maximum possible size, and its leading coefficient C° counts the maximum subsets. Theorem 5.1 then adds the fixed contribution $w(\mu)$ to the distance. □

6 Hamiltonian paths and cycles

This section addresses Question 7.5. A Hamiltonian path in these graphs is exactly a 1-change Gray code for the corresponding family of strings. Ordinary Fibonacci cubes are known to admit Hamiltonian paths [3, 16]; Fibonacci cubes with an even number of vertices, except the two-vertex case, have Hamiltonian cycles [16]. The following statements give explicit obstructions and exact finite certificates for the general p-case.

Because Γ_n^p and Λ_n^p are induced subgraphs of hypercubes, they are bipartite with parts determined by the parity of the weight. Put

$$\Delta_\Gamma^p(n) = \#\{\alpha \in \mathcal{F}_n^p : w(\alpha) \text{ even}\} - \#\{\alpha \in \mathcal{F}_n^p : w(\alpha) \text{ odd}\},$$

and define $\Delta_\Lambda^p(n)$ similarly. These quantities are useful because a path or cycle in a bipartite graph must alternate between the two parity classes.

Proposition 6.1 (Bipartition obstructions)**.** *For $n \ge 0$,*

$$\Delta_\Gamma^p(n) = \sum_{m\ge 0} (-1)^m \binom{n+p-pm}{m}.$$

For $n \ge 1$,

$$\Delta_\Lambda^p(n) = \begin{cases} 1, & 1 \le n \le p, \\ 1 + \displaystyle\sum_{1\le m\le \lfloor n/(p+1)\rfloor} (-1)^m \frac{n}{n-pm}\binom{n-pm}{m}, & n > p. \end{cases}$$

Moreover,

$$\Gamma_n^p \text{ or } \Lambda_n^p \text{ has a Hamiltonian cycle} \quad \Longrightarrow \quad \Delta = 0,$$

and

$$\Gamma_n^p \text{ or } \Lambda_n^p \text{ has a Hamiltonian path} \quad \Longrightarrow \quad |\Delta| \le 1.$$

Finally,

$$\Delta_\Gamma^p(0) = 1, \qquad \Delta_\Gamma^p(n) = 1 - n \quad (1 \le n \le p),$$

and

$$\Delta_\Gamma^p(n) = \Delta_\Gamma^p(n-1) - \Delta_\Gamma^p(n-p-1) \qquad (n \ge p+1).$$

Proof. We first count linear strings with a fixed number m of ones. If the ones are in positions

$$1 \le i_1 < i_2 < \cdots < i_m \le n,$$

the condition is $i_{j+1} - i_j \ge p+1$. Define new positions

$$j_r = i_r - p(r-1) \qquad (1 \le r \le m).$$

Then

$$1 \le j_1 < j_2 < \cdots < j_m \le n - p(m-1),$$

and this transformation is reversible. Hence the number of Fibonacci p-strings of length n with m ones is

$$\binom{n - p(m-1)}{m} = \binom{n + p - pm}{m}.$$

The difference between the even-weight and odd-weight counts is therefore obtained by alternating over m:

$$\Delta_\Gamma^p(n) = \sum_{m \ge 0} (-1)^m \binom{n + p - pm}{m}.$$

Now count cyclic strings. If $1 \le n \le p$, then $\Lambda_n^p \cong K_1$, so $\Delta_\Lambda^p(n) = 1$. Assume $n > p$ and fix $m \ge 1$. Choose a distinguished one and write the cyclic gaps after the successive ones as

$$h_1, \ldots, h_m, h_j \ge p, \qquad \sum_{j=1}^m h_j = n - m.$$

Put $r_j = h_j - p \ge 0$. Then

$$\sum_{j=1}^m r_j = n - m - pm = n - (p+1)m.$$

The number of ordered nonnegative m-tuples $(r_1, \ldots, r_m)$ with this sum is

$$\binom{n - pm - 1}{m - 1}$$

by stars and bars. There are n choices for the location of the distinguished one, and each actual string with m ones is counted m times. Therefore the number of Lucas p-strings with m ones is

$$\frac{n}{m}\binom{n - pm - 1}{m - 1} = \frac{n}{n - pm}\binom{n - pm}{m}.$$

Alternating by the parity of m and adding the empty string gives the displayed formula for $\Delta_\Lambda^p(n)$.

A Hamiltonian cycle in a bipartite graph alternates between the two bipartition classes and returns to its starting class, so it uses the same number of vertices from the two classes. Hence a Hamiltonian cycle implies $\Delta = 0$. A Hamiltonian path also alternates, but its two end vertices may lie in the same class; consequently the two class sizes may differ by at most one, so $|\Delta| \le 1$.

For Γ_n^p, the initial values are immediate. If $n = 0$, there is only the empty string. If $1 \le n \le p$, the only possible strings are the all-zero string and the n strings with a single 1, giving $\Delta_\Gamma^p(n) = 1 - n$. For $n \ge p + 1$, use the standard decomposition

$$\mathcal{F}_n^p = 0\mathcal{F}_{n-1}^p \ \cup \ 10^p\mathcal{F}_{n-p-1}^p.$$

The first part preserves weight parity, while the second part changes weight parity because of the initial 1. Therefore

$$\Delta_\Gamma^p(n) = \Delta_\Gamma^p(n-1) - \Delta_\Gamma^p(n-p-1).$$

□

The next theorem is the exact finite certificate used in computations of Hamiltonicity. It is standard in Hamiltonian graph theory; see, for example, Bondy and Murty [1]. It is useful here because all data in it are explicit from the string definition of Γ_n^p and Λ_n^p.

Theorem 6.2 (Exact Hamiltonian certificate)**.** *Let G be either Γ_n^p or Λ_n^p, and let $E(G)$ be its edge set. Then G has a Hamiltonian cycle if and only if there are variables $x_e \in \{0, 1\}$, $e \in E(G)$, satisfying*

$$\sum_{e \ni v} x_e = 2 \qquad (v \in V(G)),$$

and

$$\sum_{e \in E(G):\ e \subseteq S} x_e \le |S| - 1 \qquad (\varnothing \ne S \subsetneq V(G)).$$

If $|V(G)| \ge 2$, then G has a Hamiltonian path if and only if the graph obtained from G by adding one new vertex adjacent to every vertex of G satisfies the same Hamiltonian-cycle certificate. If $|V(G)| = 1$, then G has the trivial Hamiltonian path.

Proof. Think of $x_e = 1$ as saying that the edge e is selected. The equations

$$\sum_{e \ni v} x_e = 2 \qquad (v \in V(G))$$

say that every vertex has selected degree 2. Hence the selected edges form a spanning 2-regular subgraph, that is, a disjoint union of cycles covering all vertices. The inequalities

$$\sum_{e \subseteq S} x_e \le |S| - 1 \qquad (\varnothing \ne S \subsetneq V(G))$$

forbid any proper subset S from containing a whole selected cycle, since a cycle on $|S|$ vertices would have $|S|$ selected edges inside S. Thus the selected 2-regular spanning subgraph cannot have more than one component; it must be one Hamiltonian cycle. Conversely, the edge set of a Hamiltonian cycle plainly satisfies the degree equations and the subtour-elimination inequalities.

For the path assertion, add an apex vertex ω adjacent to every vertex of G. When $|V(G)| \ge 2$, a Hamiltonian path

$$v_1, v_2, \ldots, v_N$$

in G gives the Hamiltonian cycle

$$\omega v_1 v_2 \cdots v_N \omega$$

in the apex graph. Conversely, deleting ω from any Hamiltonian cycle of the apex graph leaves a Hamiltonian path in G. The one-vertex case is the stated trivial exception. □

Corollary 6.3 (Small ranges)**.** *For $n \le p+1$, $\Gamma_n^p \cong S_n = K_{1,n}$, with $\Gamma_0^p \cong K_1$. Hence Γ_n^p has a Hamiltonian path exactly for $n = 0,1,2$ in this range, and has no Hamiltonian cycle. For Lucas p-cubes, $\Lambda_n^p \cong K_1$ when $n \le p$ and $\Lambda_n^p \cong S_n$ when $p+1 \le n \le 2p+1$; the same star-graph conclusion applies in the latter range.*

In addition, using the isomorphisms $\Lambda_{2p+2}^p \cong Fr(4, p+1)$ and $\Lambda_{2p+3}^p \cong G_{2p+3}$ from [15, Proposition 3.7], the friendship case has a Hamiltonian path only when $p = 1$ and has no Hamiltonian cycle for all $p \ge 1$. The gear graph case has a Hamiltonian path for all $p \ge 1$ and has no Hamiltonian cycle.

Proof. The star assertions follow from [15, Propositions 2.4 and 2.5]. A star $K_{1,n}$ has a Hamiltonian path exactly when it has at most two leaves, that is, when $n \le 2$; otherwise a Hamiltonian path would have to visit more than two leaves, but every leaf can occur only as an endpoint. A star has no Hamiltonian cycle because every leaf has degree 1.

For $Fr(4, p+1)$, removing the common vertex leaves $p+1$ connected components. If a graph has a Hamiltonian path, then deleting any vertex from that path leaves at most two path components. Hence a Hamiltonian path in $Fr(4, p+1)$ can exist only when $p+1 \le 2$, that is, $p = 1$. When $p = 1$, the graph is two 4-cycles sharing one vertex, and one obtains a Hamiltonian path by starting at a noncommon vertex in one square, passing through the common vertex, and ending at a noncommon vertex in the other square. A Hamiltonian cycle cannot exist in a graph with a cut vertex unless the graph is itself a single cycle, so $Fr(4, p+1)$ has no Hamiltonian cycle.

For the gear graph G_m with $m = 2p+3$, take the hub, then an adjacent old outer vertex, and then traverse the subdivided outer cycle through all its remaining vertices. This gives a Hamiltonian path. There is no Hamiltonian cycle because the graph is bipartite with parts of different sizes: one part consists of the old outer vertices, and the other consists of the hub together with the inserted subdivision vertices. ☐

Remark 6.4. The formulas of theorem 6.1 are often the fastest way to rule out Hamiltonicity. Theorem 6.2 is a necessary-and-sufficient finite certificate for the remaining cases. It is deliberately stated as a certificate rather than as a compact congruence classification, because the bipartition sequence $\Delta_\Gamma^p(n)$ is governed by the signed recurrence $\Delta(n) = \Delta(n-1) - \Delta(n-p-1)$ and is not periodic for $p \ge 2$.

7 Concluding comments

The coefficient, root, transfer-matrix, and dynamic-programming formulas above are finite and exact. They also provide independent checks on known special cases: setting $p = 1$ in theorem 2.1 recovers the gap form of the degree sequence of Fibonacci cubes; setting $p = 1$ in theorem 3.3 gives the usual two-root formula for Fibonacci numbers; and theorem 4.1 gives c_2 of the cube polynomial when $k = 2$, in line with the cube-polynomial viewpoint of [11] and Mollard's recent proof of the p-cube cube-polynomial conjecture [13].

For computation, the degree formulas reduce the problem to extracting coefficients from rational functions; the Wiener formula reduces the coordinate sum to finitely many root terms; the induced-cycle formula uses a finite matrix of size $(p+1)^{2k}$; and the farthest-vertex formulas are dynamic programs. Thus each answer can be evaluated exactly for fixed parameters p, n, k.

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Declaration of generative AI and AI-assisted technologies in the writing process

We used ChatGPT to brainstorm the structure, anticipate potential objections, and refine the text. We are fully responsible for all claims, citations, calculations, and the final wording.

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